

The treasure-hunter



We have data,
we can visualise
it...

What's next?



Bottom-up: from data to message

Knowing your data:

- types
- values
- sparsity
- Volume
- Relations
- Contextual information

Will help you **play with its possibilities.**



Top-down: from audience to message

Knowing your audience will help you

- Define your message
- Decide
 - Which data is relevant
 - What to highlight
 - How to aggregate
 - Precision
- Decide if you need to complement available data



Decision points

Select



- What data is relevant?
- Which part of?
- Which relations?
- Which details?

Aggregate



- Shall I group the data?
- Shall I create bins?
- Which ones?

Precision

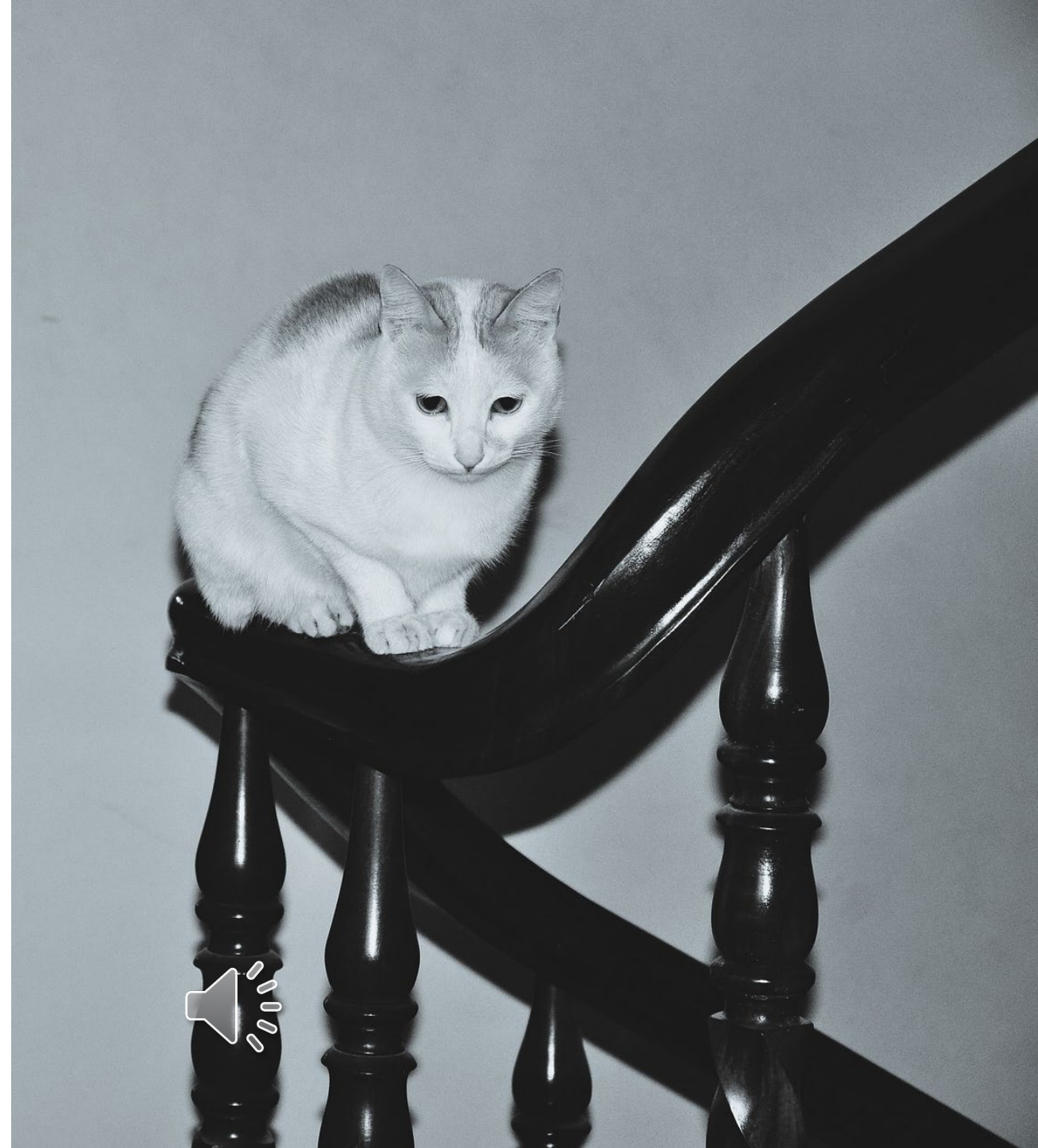


- Level of detail?
- Absolute or percentages?
- Units?
- Precision?

Which one?

- It's not Bottom-Up **OR** Top-Down
 - It's both of them!
- ...in an iterative way

Till you reach the treasure





**Today we will see the bottom-up approach
with the types of data**

**We will see both bottom-up and top-down approaches
with the types of charts**



Types of data

There is strength in numbers, but organizing those numbers is one of the great challenges.

**John C. Mather,
astrophysicist**

Objectives

- 1 Know the different types of data.
- 2 Know how to distinguish between absolute distances and distances by convention between values.
- 3 Know how to distinguish between absolute zero values and zero values by convention.
- 4 Understand the different types of order and their relationship to data types.
- 5 Have informed criteria for deciding how to visually encode colour data

Data types



Categorical

They classify data
into groups



Numerical

They report a magnitude

Data types



Categorical

They classify data
into groups



Nominal

No order



Ordinal

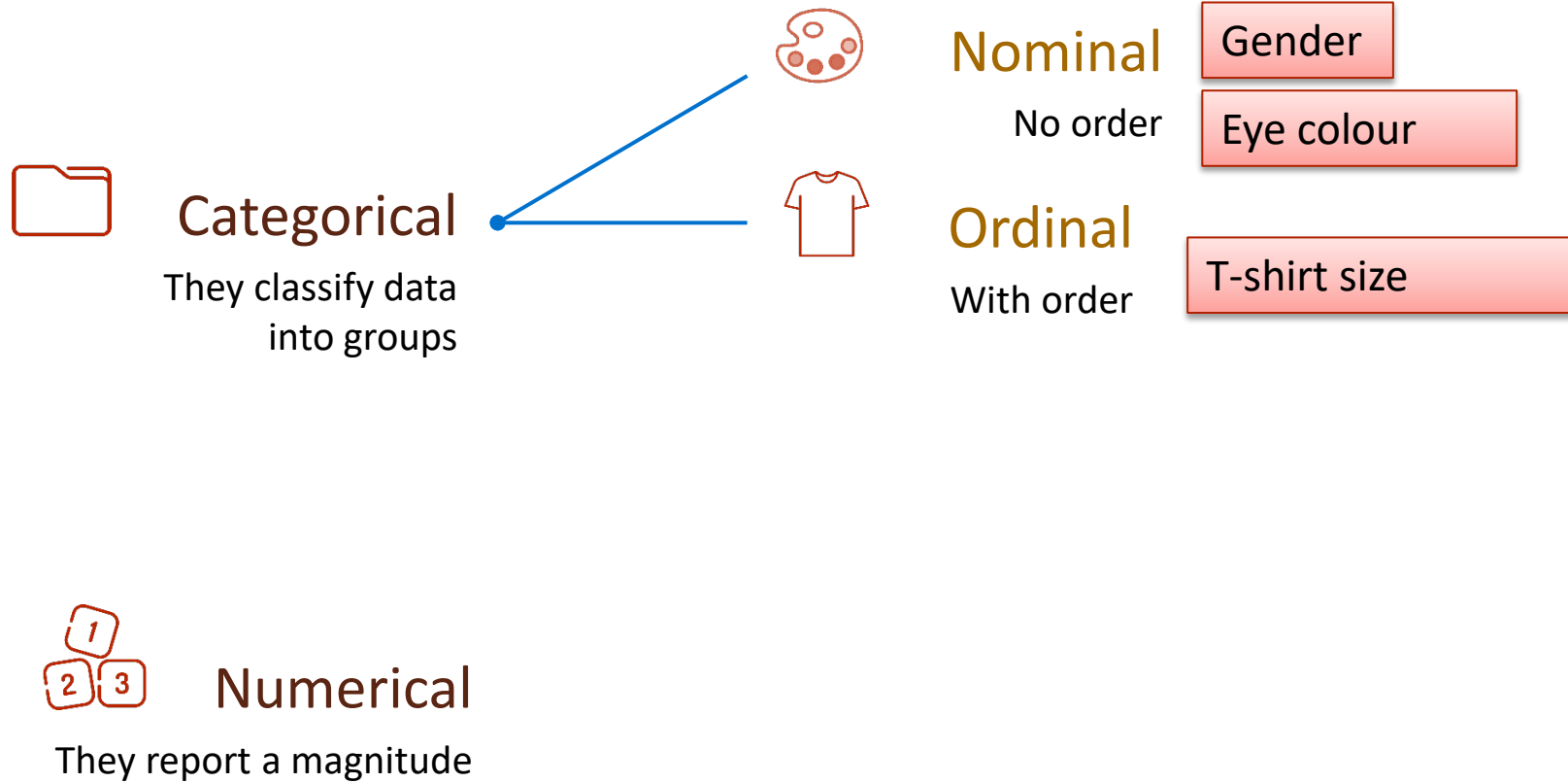
With order



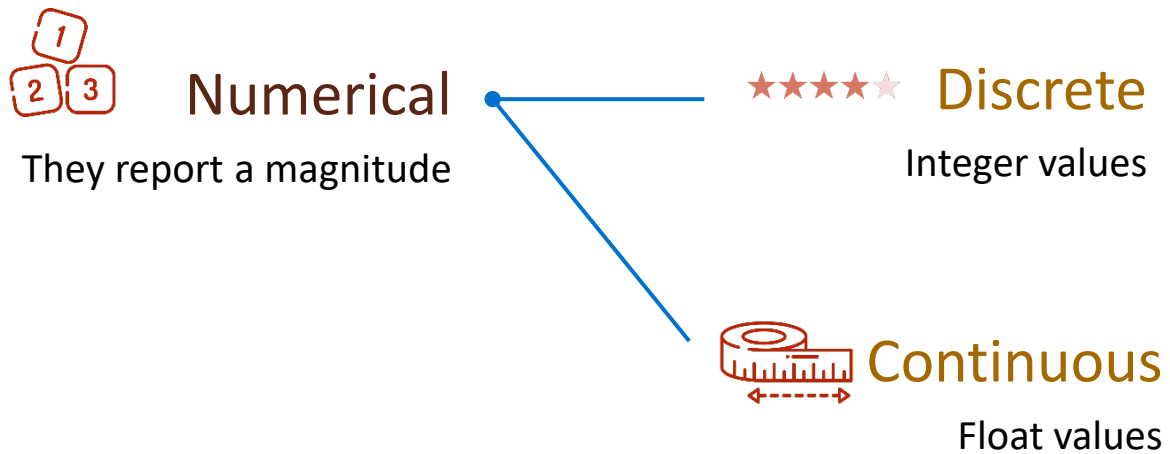
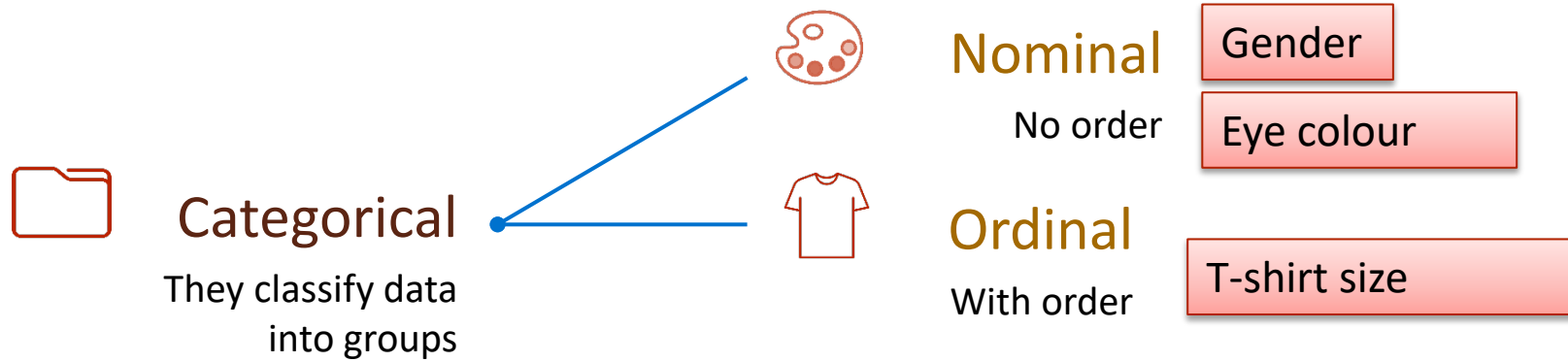
Numerical

They report a magnitude

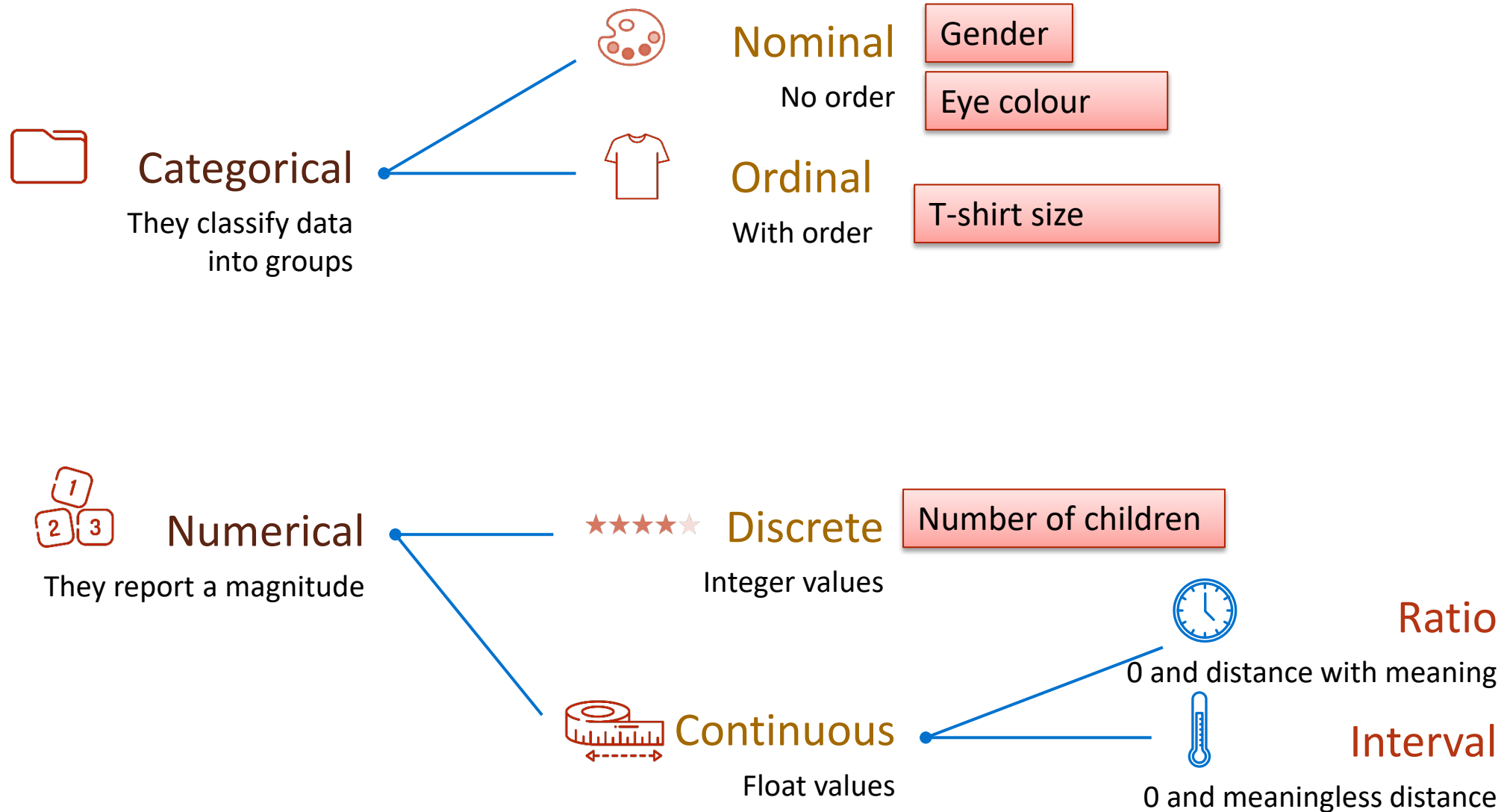
Data types



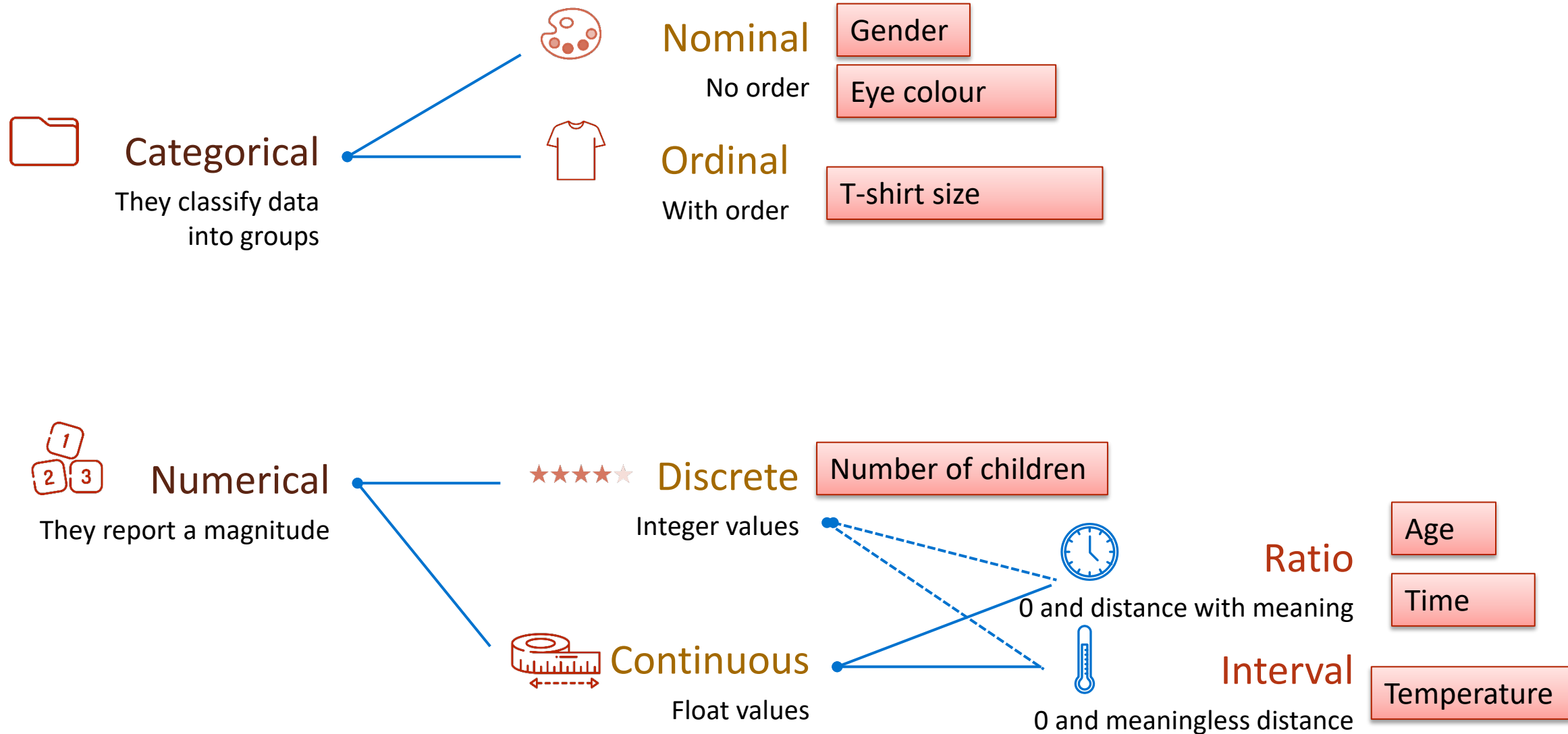
Data types



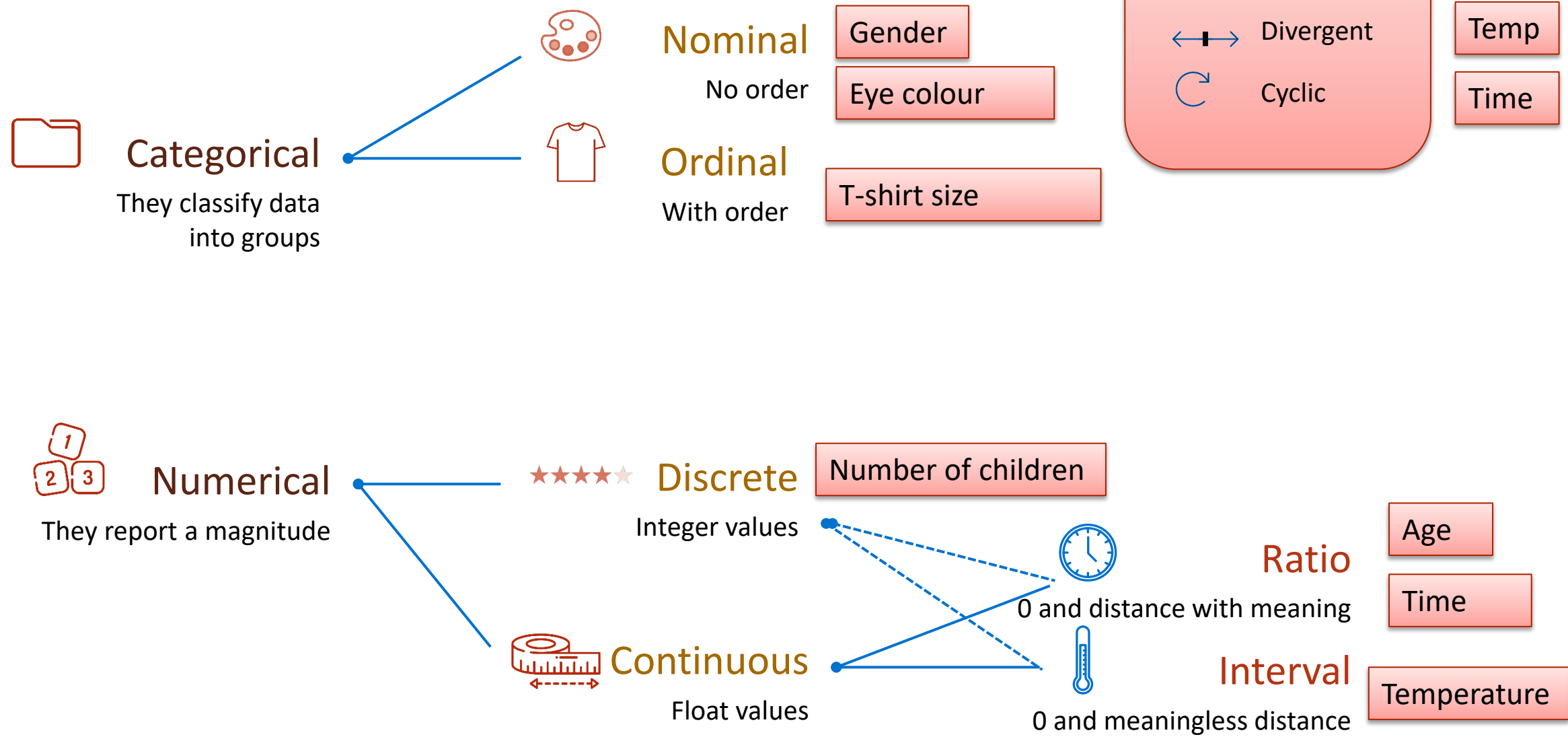
Data types



Data types



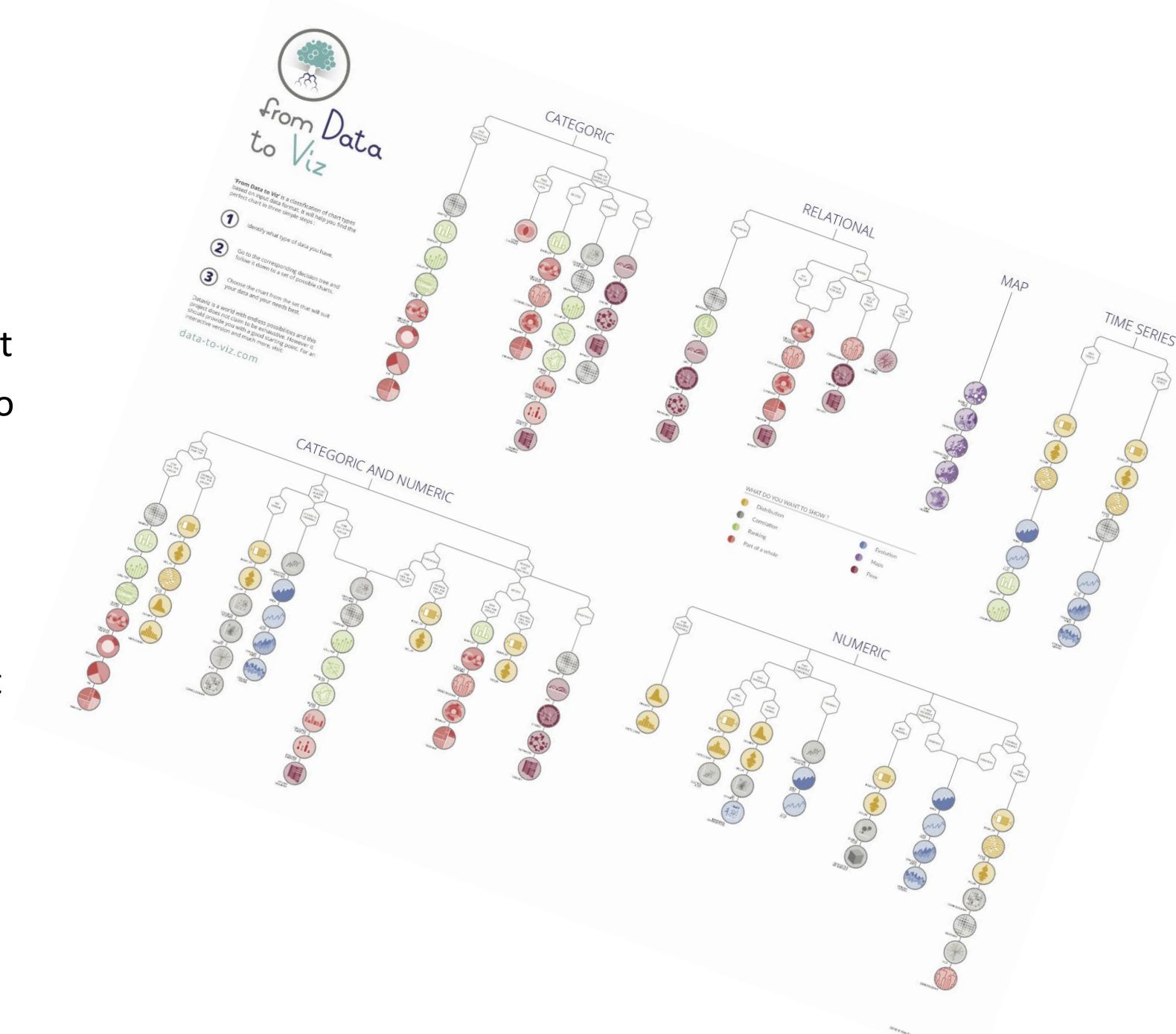
Data types



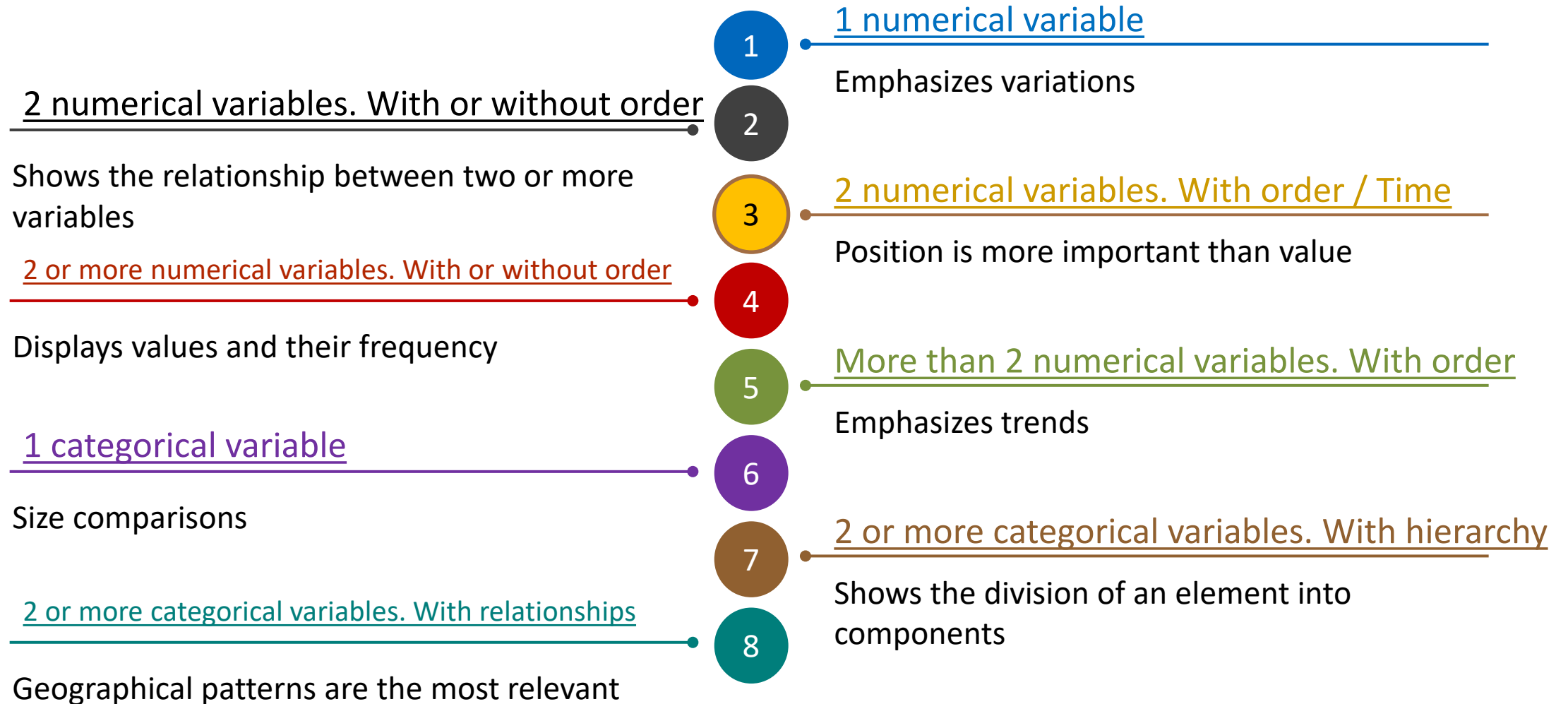
Charts by data type

In the bottom-up approach we start from the data and its possibilities to start exploring different types of graphs and encodings.

This section is based on the project classification “from Data to Viz”



Overview (data)



1

1 numerical variable

Histogram

The variable is grouped into bins and the chart displays the number of observations.



Density diagram

Displays the number of observations without groupings.



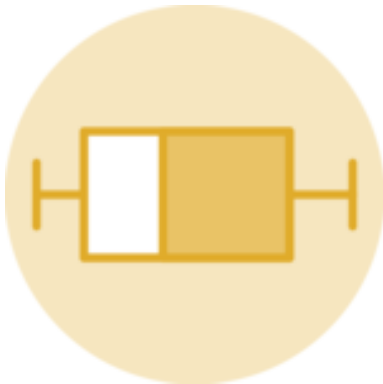
In these graphs we play with the values of the variable (x-axis) and with the count of observations (y-axis). They are based on the property of length.

2

2 numerical variables. With or Without order

Boxplot

It shows the distribution and basic descriptive statistics: median, Q1, Q3 and outliers.



Violin chart

Similar to the boxplot, it is ideal if there are many observations.



Scatter plot

Relates 2 variables.
The X-axis can be categorical.



In boxplot and violin we play with the values of the variable and with the count of observations, its design allows to compare 2 or more variables.
In the scatter plot we show the relationship between 2 variables. They are based on the position property.

3

2 numerical variables. With order / Time

Line chart

Shows the evolution of numerical values.



Area chart

Shows the evolution of numerical values.
Reports on the volume of the value y.
The y-axis must start at 0.



Both graphs allow you to combine observations of 2 or more numerical variables, facilitating their comparison.
They are based on the principle of continuity.

4

2 or more numerical variables. With or without order

Bubble chart

Adds a third dimension to the scatter plot with the size of the points.



Heat map

Adds a third dimension to the scatter plot with the color of the points.



These graphs allow you to display 3 variables with the preattentive properties of position, color and size.

5

More than 2 numerical variables. With order

Stacked area chart

Visualizes several variables, one on top of the other. Differences are perceived, not absolute values.



Stream graph

Variation that emphasizes evolution



The x-axis reflects the order in one of the numerical variables and the y-axis its magnitude. They allow the comparison of several variables. They are based on the principle of continuity.

6

1 categorical variable

Word Cloud

Displays the frequency of each word with size or color saturation.



In this graph we play with the values of the variable (words) and with the count of observations (size and/or saturation).

7

2 or more categorical variables. With hierarchy

Tree map

Categories are displayed as overlapping rectangles. The size reflects a numeric attribute.



Dendrogram

Visually displays hierarchy



In these graphs the emphasis is on the partition or grouping relationship. They use the principles of closure and the principle of connection.

8

2 or more categorical variables. With relationships

Network or Graph

Shows the interconnections between various entities.



Sankey diagram

Shows transitions through key steps.



Each categorical variable is represented by an entity, its numerical attributes are encoded with size, length... The emphasis is on their relationship. They use the principle of continuity.

Thank you and Rewrap



UNIVERSITAT DE
BARCELONA

Key ideas for reviewing the data

- 1 The selection of a chart depends on the type of data.
- 2 Although formally two values are numerically encoded in a similar way, their meaning can be very different. Graphical representation should represent meaning.
- 3 Data types are key to deciding how to operate or aggregate our values.

How to approach the visualisation:



TOBOGGAN

TOpdown

BOttomUp

and

aGAIN